

JKBOSE 2026

CLASS 11TH

Bhaiichara
ON
TOP

#XUV 

ORGANIC CHEMISTRY:

SOME BASIC PRINCIPLES AND TECHNIQUES

LECTURE-03

CHEMISTRY

VISHAL SIR



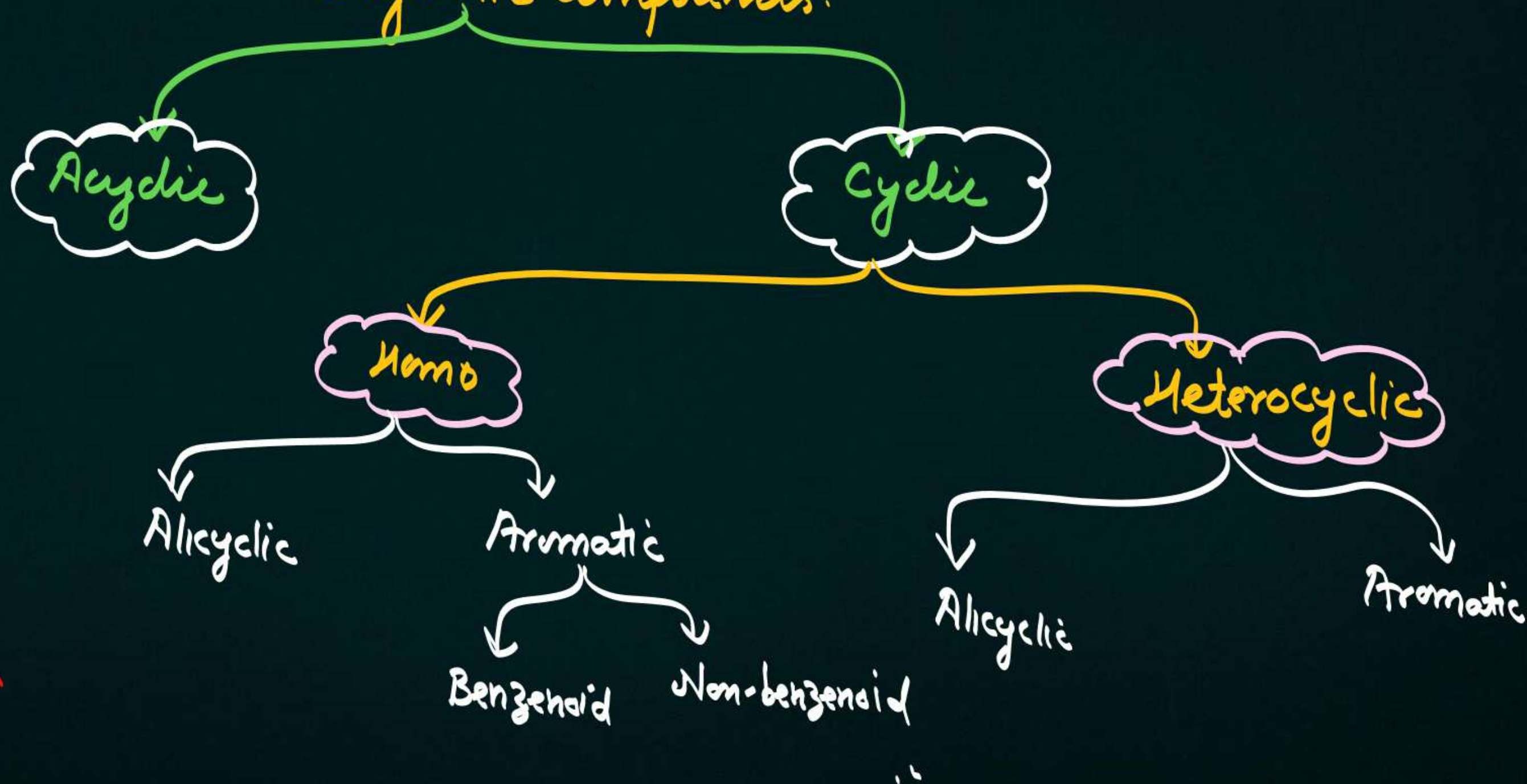
JKBose Wallah

Topics To Covered

- 1** Homologous Series
- 2** Functional Group
- 3** IUPAC Nomenclature

REVISION

Organic compounds.



FUNCTIONAL GROUP.

It is an atom or group of atom present in a organic molecule which majorly determines the chemical properties of that molecules.

e.g Carboxylic acid ($-\text{COOH}$),
alcohol ($-\text{OH}$), aldehyde ($-\text{CHO}$).

HOMOLOGOUS SERIES.

It is defined as family or series of organic compounds that show similar chemical properties and have same functional group."

Characteristics. $\Rightarrow$

- i) All the members of homologous series have same general formula.
- ii) All the members of " " have similar methods of preparation.
- iii) Adjacent members of homologous series differ by 14 amu mass or $-\text{CH}_2$ unit.
- iv) All the members have same functional group and show similar chemical properties.

e.g ALKANES. (C-C)

General formulae $\Rightarrow$ $C_n H_{2n+2}$.

$n = 1, 2, 3$ & so on

- a) $C H_4$
- b) $C_2 H_6$
- c) $C_3 H_8$
- d) $C_4 H_{10}$
- e) $C_5 H_{12}$

ALKENES. (C=C)

General formula $\Rightarrow$ $C_n H_{2n}$.

$n = 2, 3, 4$ & so on.

- a) $C_2 H_4$
- b) $C_3 H_6$
- c) $C_4 H_8$
- d) $C_5 H_{10}$
- e) $C_6 H_{12}$

ALKYNES. (C $\equiv$ C)

General formula $\Rightarrow$ $C_n H_{2n-2}$.

$n = 2, 3, 4$ & so on.

- a) $C_2 H_2$
- b) $C_3 H_4$
- c) $C_4 H_6$
- d) $C_5 H_8$
- e) $C_6 H_{10}$

IUPAC Nomenclature



IUPAC means International Union of pure and applied Chemistry.
IUPAC Nomenclature has five parts.

Secondary prefix + Primary Prefix + Word Root + Primary Suffix + Secondary Suffix.
(2° suffix) (1° prefix) (1° suffix) (2° suffix).

PREFIX

Suffix.

Name is given in a same order written above.

2° Prefix + 1° Prefix + Word Root + 1° Suffix. + 2° suffix.

↓	↓	↓	↓
SUBSTITUENT.	Cyclo	No. of C-atoms in PCC.	-ane (-) + FG. - ^{or} ene (=) - ^{or} yne (≡).

Substituent + Cyclo + WR + ^{or}ene
^{or}yne + FG.

IUPAC Nomenclature

1). Secondary Prefix $\rightarrow$ It represents the type of **SUBSTITUENT** present on a parent carbon chain.



Substituent	Prefix	Substituent	Prefix
-F	Fluoro	-NO	Nitroso
-Cl	Chloro	-N = N -	Diazo
-Br	Bromo	-OCH ₃	Methoxy
-I	Iodo	-OC ₂ H ₅	Ethoxy
-NO ₂	Nitro		
-NH ₂	Amino		

-CH₃ = Methyl.

-CH₂-CH₃ = Ethyl.
(-C₂H₅)

-CH₂-CH₂-CH₃ = Propyl

-O-CH₃ = Methoxy.

-O-C₂H₅ = Ethoxy.

-O-C₃H₇ = Propoxy
& so on.

IUPAC Nomenclature



2) PRIMARY PREFIX → It is only used for cyclic compounds. If compound is cyclic then - 1° - Prefix used. **CYCLO**

IUPAC Nomenclature

3) WORD ROOT: → It represents the no. of carbon atoms present in a parent carbon chain.



No. of C	Root	No. of C	Root
1	Meth-	15	Pentadec-
2	Eth-	16	Hexadec-
3	Prop-	17	Heptadec-
4	But-	18	Octadec-
5	Pent-	19	Nonadec-
6	Hex-	20	Icos-
7	Hept-	30	Triacont-
8	Oct-	40	Tetracont-

IUPAC Nomenclature



No. of C	Root	No. of C	Root
9	Non-	50	Pentacont-
10	Dec-	60	Hexacont-
11	Undec-	70	Heptacont-
12	Dodec-	80	Octacont-
13	Tridec-	90	Nonacont-
14	Tetradec-	100	Hect-

IUPAC Nomenclature



4) Primary Suffix $\rightarrow$ It represents whether the parent carbon chain is saturated (C-C) or Unsaturated (C=C or C $\equiv$ C).
1^o suffix

C-C	-ane
C=C	-ene
C $\equiv$ C	-yne

5). Secondary Suffix $\Rightarrow$ It represents the type of functional group present in a parent carbon chain.

IUPAC Nomenclature

1. Priority Order For Deciding Parent Chain

- (a) Maximum number of functional groups.
- (b) Maximum number of multiple bonds.
- (c) Maximum number of carbon atoms in main chain.
- (d) Maximum number of substituents.
- (e) Lowest set of locant (FG > $c-c$ multiple bonds > substituents).
- (f) Alphabetical order of name of substitute.

Naming →



i) Decide the Parent carbon chain



ii) Numbering of PCC



Naming.

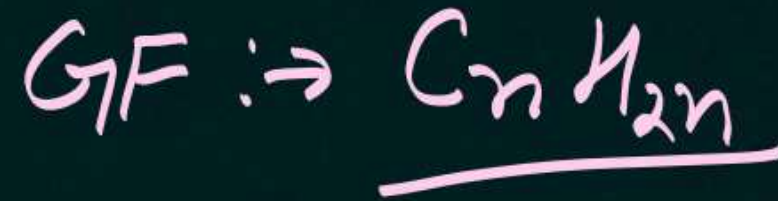
2. Priority Order For Numbering of Parent Chain

- (a) Lowest possible number is given to the functional group.
- (b) Lowest possible number is given to carbon multiple Bond.
- (c) Lowest possible number is given to the substituent
- (d) Lowest possible number is given according to alphabetical order of substituents.

QUESTION

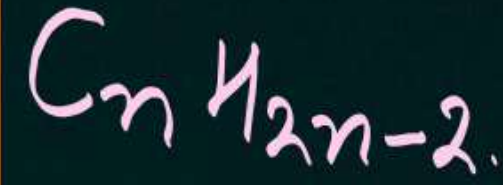
Which of the following represents the general formula of alkenes?

(PYQ)



QUESTION

Which of the following molecular formulae belongs to the alkyne series?



Lattu. Sawal.

QUESTION

Which of the following is an unsaturated compound?

→
 $=$ (Alkene) $\rightarrow C_n H_{2n}$
 or
 $\equiv$ (Alkyne) $\rightarrow C_n H_{2n-2}$

1 C_6H_{14}

2 C_4H_8

3 C_3H_7OH

4 CH_3OH

Lalhu

QUESTION

Write the IUPAC name of $\text{CH}_3 - \underset{\text{Cl}}{\underset{|}{\text{CH}}} - \text{CH}_2 - \underset{\text{Cl}}{\underset{|}{\text{CH}}} - \text{CH}_3$.

- 1 3, 4-Dichloro pentane.
- 2 3, 4-Chloro butane
- 3 2, 4-Dichloro butane
- 4 2, 4-Dichloro pentane

Summary



i) Saba Yousuf

ii) T A S.

iii) Usman Yousuf.

JEE JAAN LAGA DO.

Thank
You

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